

## ECON\*6000 Midterm Pack

Plan, MWG review with proofs, practice questions and exercises — in course order  
Microeconomics I, Fall 2026 (R. Kirkegaard) — Midterm Thursday October 29

Revised October 5, 2026: Chapter 2 removed; focus is Chapters 3–4, labs are supporting material

### How to use this pack

- The pack follows the course sequence. Each unit has four parts: **Read** (MWG summarised, with the proofs he can ask for and the page in the book and in `dokumen.pdf`, where scan page = book page +20), **Do** (practice questions in his format), **MWG exercises** (verified numbers, one line each), and **Check** (what you must be able to write from a blank page).
- **Scope, from the instructor: the midterm is mainly Chapters 3 and 4; Chapter 2 is out; the lab handouts are “good to know”.** Each unit is tagged CORE (Units 1, 2, 3, 6: MWG 3.B–3.G and 4.B, 4.D), SUPPORTING (Units 4–5: 3.H, 3.I) or GOOD TO KNOW (Appendix: comparative statics, Ch. 10). Note that the lab material MWG also covers is still Chapter 3: strict convexity vs strict quasiconcavity (p. 49, Midterm 2025 Q1.4), Kuhn–Tucker and the envelope theorem (3.D, Q2.1), integrability (3.H). What is only “good to know” is the labs’ extras: convexification, constraint qualification, ODE mechanics, monotone comparative statics.
- A “You are here” marker sits at the end of Unit 2, which is where the lecture notes stop (Lesson 7, expenditure function). Units 1–2 are review; Units 3–6 are pre-reading for the lectures to come; Unit 7 is the last three days; the Appendix is the “good to know” material.
- Proofs marked *reproduce* are exam material. Proofs marked *know the idea* are not worth memorising.
- Every “Do you agree ...?” part is answered the same way: **verdict in the first sentence**, then a proof or an explicit counterexample, then one precise sentence of economics. Name the assumption that does the work at each step (LNS, continuity, convexity, Walras’ law, homogeneity).
- No calculator on the exam. If an expression does not simplify by hand, you have made an algebra error. Pace: **10 minutes per sub-part**.
- Blue boxes are propositions, grey boxes definitions, orange boxes show how he has turned the result into a question, green boxes are things to do.

#### Start here (Tuesday, October 6)

If you have not yet done the quasilinear “compute everything” question (Question 2 here; it was Question 5 in the October 3 version), do it first, cold, in 40 minutes. It shows which parts of Units 1–2 need re-reading. Then Question 3 (new: homothetic and quasilinear structure), finish HW1 (due Sunday October 11, 23:59; your draft is in `HW1/Assignment1.tex`), and read Unit 3 before this week’s lectures finish Chapter 3.

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|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
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## The exam and the plan

### Facts

- Thursday October 29, in class, 20% of the grade. No calculator. Past class average about 35%; your target is 65%.
- **Not examinable:** Chapter 2 (instructor's confirmation, October 5). Chapter 6 is lectured on October 26–27 and is assumed not examinable.
- Format of the 2025 paper: 2 questions  $\times$  50%, each with 4 sub-parts, roughly 12.5% per sub-part.
- Each question pairs a concrete functional form with the general theorem behind it: compute first, then “Do you agree that ... for *any* preferences with property X? Prove and explain.” Some statements are true, some false.
- The mechanical parts (demand,  $v$ , Roy's identity) are about 40% of the paper. To reach 65% you need those clean and two of the four proof parts.
- He recycles homework onto exams: Midterm 2025 Q1.4 is HW1 2026 Q4 word for word. HW2 (due October 25, four days before the midterm) will telegraph Q2.
- Written precision is graded. He states assumptions explicitly and expects you to say which one does the work.
- He likes utilities that punish Lagrangian autopilot:  $(x_1 - 1)(x_2 - 1)$ ,  $x_1x_2 + x_1^2$ ,  $(\sqrt{x_1} + \sqrt{x_2})^2$ . Check convexity and corners before writing a Lagrangian.
- The envelope theorem appears on both 2025 papers (Roy, planner's value function, concavity of  $e$ ).
- With Chapter 2 gone, the 2025 midterm is the template almost exactly: Q1 on Chapter 3 (UMP,  $v$ , convexity), Q2 on the envelope theorem and Chapter 4.

### Anatomy of Midterm 2025 (the template)

Five of eight parts are Chapter 3, three are Chapter 4; nothing from Chapters 2 or 6. Expect the same shape.

| Part | Asks                                                                     | MWG        | Pack                        |
|------|--------------------------------------------------------------------------|------------|-----------------------------|
| 1.1  | $x(p, w)$ and $v(p, w)$ for $(\sqrt{x_1} + \sqrt{x_2})^2$                | 3.D        | Unit 2, Drill A             |
| 1.2  | rank $v$ at $p^1, p^2, p^3$                                              | 3.D        | Unit 2                      |
| 1.3  | $p^2$ cannot beat both $p^1$ and $p^3$ for any LNS continuous $\succsim$ | 3.D.3(iii) | Unit 2, §2.1.2              |
| 1.4  | strictly convex $\iff$ strictly quasiconcave; separable $u$              | 3.B–3.C    | Unit 1, Question 1          |
| 2.1  | Roy's identity by the envelope theorem; interpret $\lambda$              | 3.G.4      | Unit 3, Question 4          |
| 2.2  | planner's wealth-distribution problem; FOC; envelope in $p_j, w$         | 4.D        | Unit 6, Question 7          |
| 2.3  | representative consumer's demand = aggregate demand                      | 4.D.1      | Unit 6, Question 7          |
| 2.4  | Gorman form, symmetric $W$ : equal utilities, optimal rule               | 4.B, 4.D   | Unit 6, Question 7, Drill B |

**Roadmap: October 6 to October 29**

| Dates                      | Course                                                                     | This pack (time share)                                                                                                                                   | Deliverable                           |
|----------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Tue Oct 6 –<br>Sun Oct 11  | Week 5: lectures finishing<br>Chapter 3 (3.G–3.I)                          | Units 1–3 (CORE, 70%): Questions 1–4 and Drill A. Units 4–5 (SUPPORTING, 30%): one pass each, Question 5, Question 6.1–6.3.                              | <b>HW1 due Sun Oct 11</b>             |
| Mon Oct 12 –<br>Sun Oct 18 | Week 6: Thanksgiving; only<br>Thu Oct 15 lecture (aggregate demand, Ch. 4) | Unit 6 (CORE) before Thursday: Question 7 and Drill B; exercises 4.B.1, 4.D.2, 4.D.7. Then checklist rows 1–12 from memory.                              | Checklist rows 1–12 from a blank page |
| Mon Oct 19 –<br>Sun Oct 25 | Week 7: comparative statics<br>lab; aggregate demand and welfare lectures  | Unit 6 second pass against the lectures; Question 6.4–6.5; Appendix read once. HW2: send me the questions the day they are posted; mock built from them. | <b>HW2 due Sun Oct 25</b>             |
| Mon Oct 26 –<br>Wed Oct 28 | Week 8: Ch. 6 lectures (not<br>examinable)                                 | Unit 7: Midterm 2025 closed-book in 80 minutes; mock; checklist rows 1–17 from a blank page; HW1 Q4 again.                                               | Two timed papers graded               |
| Thu Oct 29                 | <b>Midterm</b>                                                             | Exam-day protocol (Unit 7)                                                                                                                               |                                       |

**Where his questions come from**

Checked against the exercise lists in your scanned MWG. Homework borrows MWG exercises; exams take the propositions and examples MWG proves in the text and wrap them in his own functional forms; homework is then recycled onto exams.

| Past question     |    | What it tests                                                                   | MWG source                                                       | In this pack          |
|-------------------|----|---------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------|
| Midterm Q1.1      | 25 | UMP for $(\sqrt{x_1} + \sqrt{x_2})^2$ : $x$ , $v$                               | CES family, Ex. 3.C.6, 3.D.5                                     | §2.1, §2.1.1          |
| Midterm Q1.2–1.3  | 25 | $p^2 = \frac{1}{2}p^1 + \frac{1}{2}p^3$ : quasiconvexity of $v$                 | Prop. 3.D.3(iii), p. 56                                          | §2.1.2                |
| Midterm Q1.4      | 25 | strictly convex $\iff$ strictly quasiconcave; separable $u$                     | Defs. 3.B.4–3.B.5, p. 49; separable part his own (cf. Ex. 3.G.4) | §1.1.2                |
| Midterm Q2.1      | 25 | Roy via the envelope theorem; interpret $\lambda$                               | Prop. 3.G.4                                                      | §3.1.1                |
| Midterm Q2.2–2.3  | 25 | planner's problem (4.D.1), FOC (4.D.2); rep. consumer demand = aggregate demand | text p. 117–118; proof of Prop. 4.D.1                            | §6.2                  |
| Midterm Q2.4      | 25 | Gorman + symmetric $W \Rightarrow$ equal utilities; optimal rule                | Example 4.D.2, Ex. 4.D.7; the twist is his                       | §6.2.1                |
| Final 25 Q1.1–1.3 |    | convexity; $x$ , $v$ ; $e$ from $v$ ; $h$ via Shephard                          | his utility; Prop. 3.G.1                                         | §1.1.2, §2.2.1        |
| Final 25 Q1.4–1.5 |    | price change along $(+t, -t)$                                                   | Prop. 3.E.2(iii) + Shephard                                      | §2.2.2                |
| Final 25 Q2       |    | median-of-three lottery; SOSD                                                   | not in MWG (Ch. 6 topic)                                         | —                     |
| Final 25 Q3       |    | Edgeworth economy; planner distributes wealth vs. bundles                       | Ex. 4.D.1; Ch. 15–16                                             | §6.2                  |
| HW1 Q1.1–1.3      |    | Slutsky symmetry at $L = 2$ ; $L = 3$ ; SARP                                    | Ex. 2.F.11 verbatim; p. 34, Ex. 2.F.10, 2.F.15; Ex. 3.J.1        | Ch. 2: not examinable |
| HW1 Q2            |    | revealed-preference table                                                       | style of Ex. 2.F.2, 2.F.3, 3.D.7                                 | Ch. 2: not examinable |
| HW1 Q3            |    | $(x_1 - 1)(x_2 - 1)$ : LNS, monotone, convex                                    | Stone–Geary twist on Ex. 3.D.6                                   | §1.1                  |
| HW1 Q4            |    | additively separable: strictly convex?                                          | his own (cf. Ex. 3.G.4)                                          | §1.1.2                |
| Oct. 5 lab        |    | recovering preferences from demand                                              | 3.H; Ex. 3.H.6, 3.H.7                                            | §4.1                  |

## Background from Chapter 2 (not examinable; used in 3.G–3.H)

Chapter 2 builds demand without preferences. What survives into Chapter 3 is notation and two identities; everything else (WARP, the compensated law of demand, revealed-preference tables, SARP in 3.J) is out of scope. If HW2 touches any of it, tell me.

- **Walras' law**  $p \cdot x(p, w) = w$  and **homogeneity of degree zero**  $x(\alpha p, \alpha w) = x(p, w)$ . In Chapter 3 both are *theorems*: Prop. 3.D.2 derives them from LNS and from the budget set being unchanged by scaling.
- **Derivative versions** (Props. 2.E.1–2.E.3, p. 27–28): differentiate  $x(\alpha p, \alpha w) = x(p, w)$  in  $\alpha$  at  $\alpha = 1$  to get  $D_p x p + D_w x w = 0$ ; differentiate  $p \cdot x = w$  in  $p_k$  and in  $w$  to get Cournot aggregation  $p^\top D_p x + x^\top = 0^\top$  and Engel aggregation  $p \cdot D_w x = 1$  (so not all goods can be inferior).
- **Slutsky matrix**  $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$ , entries  $s_{\ell k} = \partial x_\ell / \partial p_k + (\partial x_\ell / \partial w) x_k$ : the effect of  $p_k$  on  $x_\ell$  when wealth is adjusted by  $x_k dp_k$  so the old bundle stays exactly affordable. From the derivative versions,  $S p = 0$  and  $p^\top S = 0$  (Prop. 2.F.3; two lines each).
- **What Chapter 3 adds** (Props. 3.G.2–3.G.3): with utility maximization,  $S = D_p h = D_p^2 e$ , so  $S$  is symmetric and negative semidefinite. Hence  $s_{\ell \ell} \leq 0$  and, since  $\partial x_\ell / \partial p_\ell = s_{\ell \ell} - x_\ell \partial x_\ell / \partial w$ , a Giffen good must be inferior. These are the only Chapter 2 facts the exam can reach, and it reaches them through 3.G and 3.H.

# 1 Unit 1: Preferences, utility representation, convexity

**CORE. MWG 3.B–3.C and the convexity lab (Lesson 2 handout). Your notes: Lesson 4. Status: lectured; HW1 Q3–Q4 live here.**

What he tests: which axiom buys what (LNS vs monotone vs strongly monotone; continuity for representation); convex preferences vs quasiconcave utility, with the strict versions; what is ordinal and what is cardinal; additively separable utilities. Midterm 2025 Q1.4 and Final 2025 Q1.1 are this unit.

## 1.1 Read: Chapter 3.B–3.C — preferences and utility

Definitions 3.B.1–3.B.7

[MWG p. 42–45 · dokumen.pdf p. 62–65]

$\succsim$  on  $X = \mathbb{R}_+^L$  is **rational** if complete and transitive.

- **Monotone:**  $y \gg x \Rightarrow y \succ x$ . **Strongly monotone:**  $y \geq x, y \neq x \Rightarrow y \succ x$ .
- **Locally nonsatiated (LNS):** for every  $x$  and  $\varepsilon > 0$  there is  $y$  with  $\|y - x\| \leq \varepsilon$  and  $y \succ x$ .
- **Convex:** upper contour sets  $\{y : y \succsim x\}$  are convex; equivalently  $y \succsim x, z \succsim x \Rightarrow \alpha y + (1 - \alpha)z \succsim x$ .
- **Strictly convex:**  $y \succ x, z \succ x, y \neq z \Rightarrow \alpha y + (1 - \alpha)z \succ x$  for all  $\alpha \in (0, 1)$ .
- **Homothetic:**  $x \sim y \Rightarrow \alpha x \sim \alpha y$  for all  $\alpha \geq 0$ .
- **Quasilinear in good 1:** indifference sets are parallel shifts along good 1, and good 1 is desirable.

Implications: strongly monotone  $\Rightarrow$  monotone  $\Rightarrow$  LNS. None of the converses hold.

**LNS  $\Rightarrow$  the budget binds (Walras' law at the optimum)**

If  $\succsim$  is LNS and  $x^*$  is optimal in  $B_{p,w}$ , then  $p \cdot x^* = w$ .

*Proof (reproduce).* If  $p \cdot x^* < w$ , a small enough ball around  $x^*$  lies inside the budget set (intersected with  $X$ ). LNS gives a strictly preferred  $y$  in that ball, contradicting optimality. ■

### Exam angle

HW1 Q3 utility  $(x_1 - 1)(x_2 - 1)$ : check LNS, monotonicity and convexity separately, region by region. Note that LNS is enough for Walras' law; monotonicity is not needed. Thick indifference sets kill LNS.

### 1.1.1 Utility representation

Definition 3.C.1 (Continuity)

[MWG p. 46 · dokumen.pdf p. 66]

$\succsim$  is **continuous** if it is preserved under limits: whenever  $x^n \succsim y^n$  for all  $n$  with  $x^n \rightarrow x, y^n \rightarrow y$ , then  $x \succsim y$ . Equivalently, all upper and lower contour sets are closed.

**Lexicographic preferences have no utility representation**

[MWG p. 46 · dokumen.pdf p. 66]

On  $\mathbb{R}_+^2$ :  $x \succsim y$  iff  $x_1 > y_1$ , or  $x_1 = y_1$  and  $x_2 \geq y_2$ . These are rational, strongly monotone and strictly convex, but **not continuous**, and no utility function represents them.

*Proof (know the idea).* Suppose  $u$  represents  $\succsim$ . For each  $x_1$ ,  $u(x_1, 2) > u(x_1, 1)$ , so pick a rational  $r(x_1)$  strictly between. If  $x_1 > x'_1$  then  $u(x_1, 1) > u(x'_1, 2)$ , so  $r(x_1) > r(x'_1)$ . Hence  $r$  is one-to-one from  $\mathbb{R}$  into  $\mathbb{Q}$ , impossible since  $\mathbb{R}$  is uncountable. Discontinuity:  $(1/n, 0) \succ (0, 1)$  for all  $n$ , but the limit  $(0, 0) \prec (0, 1)$ . ■

**Proposition 3.C.1**

[MWG p. 47 · dokumen.pdf p. 67]

If  $\succsim$  is rational and continuous, a *continuous* utility function represents it.

*Proof for  $X = \mathbb{R}_+^L$  and monotone  $\succsim$  (reproduce the construction).* Let  $e = (1, \dots, 1)$ . For each  $x$ , the sets  $\{\alpha \geq 0 : \alpha e \succsim x\}$  and  $\{\alpha \geq 0 : x \succsim \alpha e\}$  are nonempty (monotonicity:  $0 \preceq x$  and  $\bar{\alpha}e \succ x$  for  $\bar{\alpha} > \max_\ell x_\ell$ ) and closed (continuity), and they cover  $[0, \infty)$  (completeness). Since  $[0, \infty)$  is connected, they intersect: some  $\alpha(x)$  has  $\alpha(x)e \sim x$ . Monotonicity makes it unique. Define  $u(x) = \alpha(x)$ . Then  $x \succsim y \iff \alpha(x)e \succsim \alpha(y)e \iff \alpha(x) \geq \alpha(y)$  by transitivity and monotonicity. (Continuity of  $u$  is shown separately.) ■

**Ordinal vs. cardinal.** If  $u$  represents  $\succsim$ , so does  $f \circ u$  for any strictly increasing  $f$ . Properties that survive such transformations (quasiconcavity, monotonicity, homotheticity of preferences) are properties of  $\succsim$ . Properties that do not (concavity, additivity,  $u(\alpha x) = \alpha u(x)$ ) are properties of a particular  $u$ .

**1.1.2 Convexity vs. quasiconcavity****Quasiconcavity**

[MWG p. 49 · dokumen.pdf p. 69]

$u$  is **quasiconcave** if  $\{y : u(y) \geq u(x)\}$  is convex for all  $x$ , i.e.  $u(\alpha y + (1 - \alpha)z) \geq \min\{u(y), u(z)\}$ . **Strictly quasiconcave:**  $u(\alpha y + (1 - \alpha)z) > \min\{u(y), u(z)\}$  whenever  $y \neq z$ ,  $\alpha \in (0, 1)$ .

$\succsim$  **convex**  $\iff u$  **quasiconcave**;  $\succsim$  **strictly convex**  $\iff u$  **strictly quasiconcave**

For any  $u$  representing  $\succsim$ .

*Proof of the strict version (reproduce; this was Midterm 2025 Q1.4).* ( $\Rightarrow$ ) Take  $y \neq z$ ,  $\alpha \in (0, 1)$ . Without loss of generality  $u(y) \geq u(z)$ , so  $y \succsim z$ ; also  $z \succsim z$ . Apply strict convexity with  $x = z$ :  $\alpha y + (1 - \alpha)z \succ z$ , i.e.  $u(\alpha y + (1 - \alpha)z) > u(z) = \min\{u(y), u(z)\}$ . ( $\Leftarrow$ ) Let  $y \succsim x$ ,  $z \succsim x$ ,  $y \neq z$ . Then  $\min\{u(y), u(z)\} \geq u(x)$ , and strict quasiconcavity gives  $u(\alpha y + (1 - \alpha)z) > \min\{u(y), u(z)\} \geq u(x)$ , i.e.  $\alpha y + (1 - \alpha)z \succ x$ . The weak version is identical with  $\geq$  in place of  $>$ . ■

**Useful facts.**

- Concave  $\Rightarrow$  quasiconcave (since  $\alpha u(y) + (1 - \alpha)u(z) \geq \min\{u(y), u(z)\}$ ). Strictly concave  $\Rightarrow$  strictly quasiconcave. Converses fail:  $u = x_1 x_2$  is quasiconcave but not concave on  $\mathbb{R}_+^2$ , while  $\sqrt{x_1 x_2}$  represents the same preferences and is concave.
- Convex preferences do *not* imply every representation is concave (cardinal property). Not every convex preference has a concave representation, but you will not need that subtlety.
- A sum of concave functions is concave; it is strictly concave if, for every direction  $y - z \neq 0$ , at least one summand is strictly concave along that direction. Use this directly on additively separable utilities  $u = \sum_i f_i(x_i)$  and ask: along which directions  $y - z$  can *every* summand be linear? (That is exactly what HW1 Q4.2 tests.)
- The **strict** inequality is where marks are lost. Always check whether the direction you are moving in makes the inequality strict.

**1.2 Do: Question 1 — Preferences, convexity and utility representation**

**Covers:** MWG Chapter 3 (3.B–3.C); Convexity lab (quasiconcavity, existence/uniqueness).

*Mimics:* Midterm 2025 Q1.4; HW1 Q3–Q4; Final 2025 Q1.1.

- 1.1** Do you agree that any preference relation on  $\mathbb{R}_+^2$  that is complete, transitive and strongly monotone admits a utility representation? Prove or give a counterexample. What assumption, if added, restores the claim?
- 1.2** Let  $u$  represent  $\succsim$ . Do you agree with each of the following? Prove or give a counterexample.
- If  $u$  is quasiconcave then  $\succsim$  is convex.
  - If  $\succsim$  is convex then *every* utility function representing  $\succsim$  is quasiconcave.
  - If  $\succsim$  is convex then *every* utility function representing  $\succsim$  is concave.
  - If  $\succsim$  is convex then *some* utility function representing  $\succsim$  is concave.
- 1.3** Consider  $u(x_1, x_2) = \max\{x_1, x_2\}$  on  $\mathbb{R}_+^2$ . Are the preferences convex? Monotone? Strongly monotone? Locally non-satiated? Continuous? Derive the Walrasian demand correspondence  $x(p, w)$  for all  $p \gg 0, w > 0$ . Is it a function? Is it convex-valued? Which hypothesis of MWG Proposition 3.D.3 fails?
- 1.4** Consider  $u(x_1, x_2) = x_1 + \min\{x_1, x_2\}$  on  $\mathbb{R}_+^2$ . Are the preferences convex? Strictly convex? Sketch the indifference curve through  $(1, 1)$ . Derive the Walrasian demand correspondence. For which price vectors is it not single-valued?
- 1.5** Let  $\succsim$  be continuous and strictly convex on  $X = \mathbb{R}_+^2$ . Do you agree that the UMP has a unique solution for every  $p \gg 0$  and  $w > 0$ ? State which theorem delivers existence and which argument delivers uniqueness. Does your answer change if  $p_1 = 0$ ? If  $X = \mathbb{R}_{++}^2$ ? Explain.

### 1.3 MWG exercises

Verified against your scanned copy; numbers follow the 1995 edition. \* = closest to his style.

- **3.B.2** transitive + LNS + weakly monotone  $\Rightarrow$  monotone.
- **3.B.3** draw a convex, locally nonsatiated preference that is not monotone.
- **3.C.1\*** verify that the lexicographic ordering is complete, transitive, strongly monotone and strictly convex (and still has no utility).
- **3.C.3** closed upper and lower contour sets  $\Rightarrow$  continuity in the sense of Definition 3.C.1.
- **3.C.4** a discontinuous preference that is nonetheless represented by a utility function.
- **3.C.5\*** continuous  $\succsim$  is homothetic iff some representation is HD1; quasilinear iff some representation is  $x_1 + \phi(x_2, \dots, x_L)$ ; why these are cardinal properties. **Feeds Question 3.**
- **3.C.6** CES:  $\rho = 1$  linear,  $\rho \rightarrow 0$  Cobb–Douglas,  $\rho \rightarrow -\infty$  Leontief.

### 1.4 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|   | Result                                                                  | MWG          | Seen on                    |
|---|-------------------------------------------------------------------------|--------------|----------------------------|
| 1 | LNS $\Rightarrow$ Walras' law at the optimum                            | 3.D.2, p. 51 | HW1 Q3                     |
| 2 | Lexicographic preferences: no utility representation                    | p. 46        | —                          |
| 3 | Continuous + monotone $\Rightarrow$ utility (diagonal construction)     | 3.C.1, p. 47 | —                          |
| 4 | Strictly convex $\iff$ strictly quasiconcave; sums of concave functions | p. 49        | Midterm 25 Q1.4;<br>HW1 Q4 |

## 2 Unit 2: UMP, EMP, Kuhn–Tucker, the envelope theorem

**CORE. MWG 3.D–3.E and the optimization lab (Lesson 5 handout). Your notes: Lessons 6–7. Status: lectured up to the expenditure function.**

What he tests: deriving  $x, v, h, e$  by hand for non-standard utilities, with corners; the properties of  $v$  (quasiconvex) and  $e$  (concave) and their proofs without derivatives; the duality identities; what homotheticity and quasilinearity do to all four objects. Midterm 2025 Q1.1–1.3 and Final 2025 Q1.2–1.5 are this unit.

### 2.1 Read: Chapter 3.D — the utility maximization problem

$$v(p, w) = \max_{x \geq 0} u(x) \quad \text{s.t.} \quad p \cdot x \leq w.$$

**Existence (Prop. 3.D.1, p. 50):** if  $p \gg 0$  and  $u$  is continuous, a solution exists (Weierstrass:  $B_{p,w}$  is closed and bounded, hence compact, and nonempty). If some  $p_\ell = 0$ , the budget set is unbounded and existence can fail.

**Proposition 3.D.2 (properties of  $x(p, w)$ )**

[MWG p. 51 · dokumen.pdf p. 71]

$u$  continuous,  $\succsim$  LNS on  $\mathbb{R}_+^L$ . Then (i) HD0; (ii) Walras' law; (iii)  $\succsim$  convex  $\Rightarrow x(p, w)$  is a convex set;  $\succsim$  strictly convex  $\Rightarrow x(p, w)$  is a single point.

*Proof (reproduce).* (i)  $B_{\alpha p, \alpha w} = B_{p, w}$ . (ii) LNS argument above. (iii) If  $x, x' \in x(p, w)$  then  $u(x) = u(x') = v$ . Any  $\alpha x + (1 - \alpha)x'$  is affordable (budget set convex). Convexity gives  $u(\alpha x + (1 - \alpha)x') \geq v$ , so it is optimal. Under strict convexity with  $x \neq x'$ ,  $u(\alpha x + (1 - \alpha)x') > v$ , a contradiction. ■

#### 2.1.1 Kuhn–Tucker conditions

[MWG p. 53 · dokumen.pdf p. 73]

If  $x^*$  solves the UMP and  $u$  is differentiable, there is  $\lambda \geq 0$  with

$$\frac{\partial u(x^*)}{\partial x_\ell} \leq \lambda p_\ell \quad \text{for all } \ell, \quad \text{with equality if } x_\ell^* > 0.$$

- Interior:  $\nabla u(x^*) = \lambda p$ , so  $\text{MRS}_{\ell k} = p_\ell / p_k$ .
- Corner  $x_k^* = 0$ :  $\partial u / \partial x_k \leq \lambda p_k$ , i.e. the marginal utility per dollar of  $k$  is no higher than for goods consumed.
- **Sufficiency:** if  $u$  is quasiconcave and monotone with  $\nabla u(x) \neq 0$  for all  $x$ , the KT conditions are also sufficient. Always say this sentence when you use FOCs.
- $\lambda = \partial v / \partial w$  is the **marginal utility of wealth** (envelope theorem, §3.1.1). It is not invariant to monotone transformations of  $u$ .

#### Exam angle

Quasilinear  $u = x_1 + f(x_2)$  and perfect substitutes generate corners; Leontief has a kink (no FOC). His non-standard utilities test whether you check the shape *before* writing a Lagrangian. For  $(\sqrt{x_1} + \sqrt{x_2})^2$ : monotone transform of  $\sqrt{x_1} + \sqrt{x_2}$ , so solve that (CES with  $\rho = 1/2$ ).

### 2.1.2 The indirect utility function

#### Proposition 3.D.3 (properties of $v$ )

[MWG p. 56 · dokumen.pdf p. 76]

$u$  continuous,  $\succsim$  LNS on  $\mathbb{R}_+^L$ . Then  $v(p, w)$  is (i) HD0; (ii) strictly increasing in  $w$ , nonincreasing in each  $p_\ell$ ; (iii) **quasiconvex**:  $\{(p, w) : v(p, w) \leq \bar{v}\}$  is convex for every  $\bar{v}$ ; (iv) continuous.

*Proof of (ii) and (iii) (reproduce (iii), it was Midterm 2025 Q1.3).* (ii) If  $w' > w$ ,  $x(p, w)$  is strictly inside  $B_{p, w'}$ , so LNS gives an affordable strictly better bundle. Raising  $p_\ell$  shrinks the budget set. (iii) Let  $v(p, w) \leq \bar{v}$ ,  $v(p', w') \leq \bar{v}$ , and  $(p'', w'') = \alpha(p, w) + (1 - \alpha)(p', w')$ . Take any  $x$  with  $p'' \cdot x \leq w''$ , i.e.  $\alpha p \cdot x + (1 - \alpha)p' \cdot x \leq \alpha w + (1 - \alpha)w'$ . Then  $p \cdot x \leq w$  or  $p' \cdot x \leq w'$  (otherwise both strict inequalities the other way, and summing contradicts). So  $x$  was affordable in one of the two original situations and  $u(x) \leq \bar{v}$ . Since  $x$  was arbitrary in  $B_{p'', w''}$ ,  $v(p'', w'') \leq \bar{v}$ . ■

#### Exam angle: Midterm 2025 Q1.3

$p^2 = (2, 2) = \frac{1}{2}(1, 3) + \frac{1}{2}(3, 1)$ , same  $w$ . Quasiconvexity:  $v(p^2, w) \leq \max\{v(p^1, w), v(p^3, w)\}$ . So  $p^2$  **cannot** be strictly better than both, for any LNS continuous preferences. Intuition: any bundle affordable at averaged prices was affordable at one of the extremes. Note:  $v$  is *quasiconvex*, not quasiconcave; it does not require  $u$  to be quasiconcave.

## 2.2 Read: Chapter 3.E — the expenditure minimization problem

$$e(p, u) = \min_{x \geq 0} p \cdot x \quad \text{s.t.} \quad u(x) \geq u, \quad h(p, u) = \text{argmin.}$$

#### Proposition 3.E.1 (UMP–EMP duality)

[MWG p. 58 · dokumen.pdf p. 78]

$u$  continuous,  $\succsim$  LNS,  $p \gg 0$ . (i) If  $x^*$  solves the UMP at  $w > 0$ , then  $x^*$  solves the EMP at  $u = u(x^*)$ , and  $e(p, u(x^*)) = w$ . (ii) If  $x^*$  solves the EMP at  $u > u(0)$ , then  $x^*$  solves the UMP at  $w = p \cdot x^*$ , and  $v(p, p \cdot x^*) = u$ .

*Proof (reproduce (i); know (ii)).* (i) Suppose not: some  $x'$  with  $u(x') \geq u(x^*)$  and  $p \cdot x' < p \cdot x^* \leq w$ . By LNS there is  $x''$  close to  $x'$  with  $u(x'') > u(x') \geq u(x^*)$  and still  $p \cdot x'' < w$ . This contradicts  $x^*$  solving the UMP. Expenditure is  $p \cdot x^* = w$  by Walras' law.

(ii) Suppose not: some  $x'$  with  $u(x') > u(x^*)$  and  $p \cdot x' \leq p \cdot x^*$ . Since  $u > u(0)$ ,  $x^* \neq 0$  and  $p \cdot x^* > 0$ . For  $\alpha < 1$  close to 1, continuity gives  $u(\alpha x') > u(x^*)$  while  $p \cdot \alpha x' < p \cdot x^*$ , contradicting  $x^*$  solving the EMP. ■

### 2.2.1 Duality identities

$$e(p, v(p, w)) = w, \quad v(p, e(p, u)) = u, \quad h(p, u) = x(p, e(p, u)), \quad x(p, w) = h(p, v(p, w)).$$

Final 2025 Q1.3 is this: get  $e$  by solving  $v(p, e) = u$  for  $e$ , then  $h = \nabla_p e$ .

#### Proposition 3.E.2 (properties of $e$ )

[MWG p. 59 · dokumen.pdf p. 79]

$u$  continuous,  $\succsim$  LNS.  $e(p, u)$  is (i) HD1 in  $p$ ; (ii) strictly increasing in  $u$ , nondecreasing in each  $p_\ell$ ; (iii) **concave in  $p$** ; (iv) continuous.

*Proof of (i), (iii) (reproduce (iii)).* (i) The constraint  $u(x) \geq u$  does not involve  $p$ , so minimizing  $\alpha p \cdot x$  gives the same minimizers; the value scales by  $\alpha$ .

(iii) Let  $p'' = \alpha p + (1 - \alpha)p'$  and  $x'' \in h(p'', u)$ . Then

$$e(p'', u) = p'' \cdot x'' = \alpha p \cdot x'' + (1 - \alpha)p' \cdot x'' \geq \alpha e(p, u) + (1 - \alpha)e(p', u),$$

since  $x''$  is feasible (reaches  $u$ ) at every price vector, so  $p \cdot x'' \geq e(p, u)$  and  $p' \cdot x'' \geq e(p', u)$ . ■

**Intuition:** if the consumer kept buying the old bundle, cost would be linear in  $p$ ; re-optimizing can only do better, so  $e$  lies below its tangent lines.

**Propositions 3.E.3–3.E.4 (properties of  $h$ )**

[MWG p. 61–62 · dokumen.pdf p. 81–82]

**3.E.3:**  $h$  is HD0 in  $p$ ; no excess utility ( $u(x) = u$  for  $x \in h(p, u)$ ); convex-valued if  $\succsim$  convex, single-valued if strictly convex.

**3.E.4 (compensated law of demand for  $h$ ):** if  $h$  is single-valued,  $(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0$ .

*Proof of 3.E.4 (reproduce; three lines).*

$$(p'' - p') \cdot (h'' - h') = \underbrace{(p'' \cdot h'' - p'' \cdot h')}_{\leq 0 \text{ (} h'' \text{ cheapest at } p'')} + \underbrace{(p' \cdot h' - p' \cdot h'')}_{\leq 0 \text{ (} h' \text{ cheapest at } p')} \leq 0.$$

■

Own-price version:  $\partial h_\ell / \partial p_\ell \leq 0$ . **Hicksian demand never slopes up**; Walrasian demand can.

### 2.2.2 Application: Final 2025 Q1.4–1.5 (price change along $(+t, -t)$ )

At  $p$ ,  $x_1 = x_2$ , so  $h_1(p, u) = h_2(p, u)$ . Consider  $g(t) = e(p + t(1, -1), u)$ .  $g$  is concave in  $t$  (restriction of a concave function to a line) and, by Shephard,  $g'(0) = h_1 - h_2 = 0$ . A concave function lies below its tangent, so  $g(t) \leq g(0)$ : the consumer needs **weakly less** money to reach the same utility. Intuition: at the starting bundle the cost of the old bundle is unchanged ( $tx_1 - tx_2 = 0$ ), and re-optimizing can only lower cost. This uses only LNS and continuity (plus strict convexity for single-valued  $h$ ), not the functional form.

## 2.3 Do: Question 2 — Quasilinear utility, corner solutions and duality

**Covers:** MWG Chapter 3 (3.D–3.G); Optimization lab (KKT, corner solutions, constraint qualification, envelope theorem).

*Mimics:* Midterm 2025 Q1.1–1.3; Final 2025 Q1.2–1.5. This is the “compute everything” question.

A consumer has preferences on  $\mathbb{R}_+^2$  represented by

$$u(x_1, x_2) = x_1 + 2\sqrt{x_2}.$$

- 2.1** Are the preferences convex? Strictly convex? Strongly monotone? Explain briefly.
- 2.2** Derive the Walrasian demand function  $x(p, w)$  for all  $p \gg 0$ ,  $w > 0$ , *including the corner solution*. Write down the Kuhn–Tucker conditions and explain why they are necessary and sufficient here. Under what condition on  $(p, w)$  is the solution interior?
- 2.3** For the interior case, derive the indirect utility function  $v(p, w)$ . Verify that it is homogeneous of degree zero and verify Roy’s identity for good 1.
- 2.4** For the interior case, derive the expenditure function  $e(p, u)$  by inverting  $v$ . Derive the Hicksian demands using Shephard’s lemma. Verify that  $h_2(p, u) = x_2(p, w)$ . Explain, in terms of wealth effects, why this must be so.
- 2.5** Compute  $D_p h(p, u)$ . Show it is symmetric and negative semidefinite, and that  $D_p h(p, u)p = 0$ . Do you agree that for *any* continuous and locally non-satiated preference relation (differentiable or not),  $e(\cdot, u)$  is concave in  $p$ ? Prove your answer without using derivatives and explain the economic intuition.
- 2.6** (The instructor’s twist.) Fix  $w$  and start from prices  $(p_1, p_2)$  with an interior solution. Prices change to  $(p_1 + t, p_2 - t)$  with  $t > 0$  small. Does the consumer need more or less wealth than before to reach the *original* utility level? Answer first for this specific  $u$  (compute), then for a general LNS continuous preference relation [Hint: Final 2025 Q1.5].

## 2.4 Do: Question 3 — Homothetic and quasilinear preferences: the structure of $x$ , $v$ , $h$ , $e$

**Covers:** MWG 3.C (Ex. 3.C.5), 3.D (Ex. 3.D.3, 3.D.4), 3.E (Ex. 3.E.5, 3.E.7), 3.G (Gorman form, Ex. 3.G.10–3.G.12). Parts 3.2 and 3.4 use Roy’s identity (Lesson 7).

*Mimics:* his “is this specific to ...?” pattern, and the two preference classes that produce the Gorman form on which Midterm 2025 Q2.4 rests.

Throughout,  $u$  is continuous and represents a locally nonsatiated, strictly convex preference relation;  $p \gg 0$ ,  $w > 0$ ; solutions are interior where differentiability is used.

- 3.1** Suppose  $u$  is homogeneous of degree one on  $\mathbb{R}_+^L$ . Do you agree that  $x(p, w) = w x(p, 1)$  and  $v(p, w) = w v(p, 1)$ ? Prove. [Hint: if  $x^*$  solves the UMP at  $(p, 1)$ , what does  $w x^*$  do at  $(p, w)$ ?] Conclude that  $h(p, u) = u h(p, 1)$  and  $e(p, u) = u e(p, 1)$ , and that the wealth expansion path at any  $p$  is a ray through the origin.
- 3.2** Now the converse. Assume in addition that  $u$  is differentiable and strictly quasiconcave. Do you agree that “if  $v(p, w)$  is homogeneous of degree one in  $w$ , then preferences are homothetic”? Prove or give a counterexample. [Hint: Roy’s identity makes  $x(p, w)$  linear in  $w$ ; every  $x \gg 0$  is demanded at  $p = \nabla u(x)$ ,  $w = p \cdot x$ ; evaluate  $v$  along the ray through  $x$ .]
- 3.3** Quasilinear case:  $X = \mathbb{R} \times \mathbb{R}_+$ ,  $u(x_1, x_2) = x_1 + \phi(x_2)$  with  $\phi$  strictly concave, differentiable and  $\phi' > 0$ ; normalize  $p_1 = 1$ . Do you agree that (a)  $x_2(p, w)$  does not depend on  $w$ ; (b)  $v(p, w) = w + \psi(p_2)$  for some function  $\psi$ ; (c)  $h_2(p, u)$  does not depend on  $u$ ; (d)  $e(p, u) = u - \psi(p_2)$ ? Prove each, give  $\psi$  explicitly in terms of  $\phi$  and  $x_2(p_2)$ , and verify that Shephard’s lemma applied to (d) returns the  $x_2$  of (a). What is the wealth expansion path?
- 3.4** Both classes have indirect utility of the Gorman form  $v(p, w) = a(p) + b(p)w$ . Identify  $a$  and  $b$  in each case (without the normalization  $p_1 = 1$  in the quasilinear case) and check that  $v$  is homogeneous of degree zero. Do you agree that “the wealth expansion path of a consumer at fixed  $p$  is a straight line if  $v$  is of the Gorman form”? Prove it using Roy’s identity. State what the converse claims and why it is harder (you may cite Prop. 4.B.1).
- 3.5** Concrete check: for  $u = x_1^{1/3} x_2^{2/3}$  and for  $u = x_1 + 2 \ln x_2$  (with  $p_1 = 1$  and  $w \geq 2$ ), compute  $x$ ,  $v$ ,  $e$  and  $h$ , sketch the wealth expansion paths at  $p = (1, 1)$ , and verify 3.1 and 3.3. Then: do you agree that for the quasilinear consumer, good 2 is neither normal nor inferior, and that therefore its Marshallian and Hicksian demands coincide as functions of  $p$ ? Explain using the Slutsky equation.

## 2.5 Drill A: Chapter 3 computations by hand

Six utilities, same routine each time. This is the 40% of the paper that must be automatic. No calculator; every line must close by hand.

For each utility: (i) convexity and monotonicity of the preferences, and whether a corner or a kink is possible; (ii)  $x(p, w)$ , including the corner region; (iii)  $v(p, w)$ , check homogeneity of degree zero and Walras’ law; (iv)  $e(p, u)$  by inverting  $v$ ; (v)  $h(p, u)$  by Shephard’s lemma, and check  $h(p, v(p, w)) = x(p, w)$ ; (vi) verify Roy’s identity for good 1; (vii) sign  $s_{11}$ .

1.  $u = x_1^\alpha x_2^{1-\alpha}$ ,  $\alpha \in (0, 1)$ . Checkpoint:  $v = w \alpha^\alpha (1-\alpha)^{1-\alpha} p_1^{-\alpha} p_2^{-(1-\alpha)}$ .
2.  $u = (x_1^{-1} + x_2^{-1})^{-1} = \frac{x_1 x_2}{x_1 + x_2}$  (CES with  $\rho = -1$ ; the dual twin of his  $\rho = \frac{1}{2}$  midterm utility). Checkpoint:  $v = \frac{w}{(\sqrt{p_1} + \sqrt{p_2})^2}$ , so  $e = u(\sqrt{p_1} + \sqrt{p_2})^2$ .
3.  $u = \min\{x_1, 2x_2\}$  (kink: no FOC, use the structure). Checkpoint:  $v = \frac{2w}{2p_1 + p_2}$ .
4.  $u = x_1 + 2x_2$  (corners; demand is a correspondence on one price ray). Checkpoint:  $v = \max\{w/p_1, 2w/p_2\}$ .
5.  $u = (x_1 - 1)(x_2 - 1)$  on the region  $w > p_1 + p_2$  (HW1 Q3’s utility, Stone–Geary). Checkpoint:

$$v = \frac{(w - p_1 - p_2)^2}{4p_1p_2}, \text{ so } e = p_1 + p_2 + 2\sqrt{up_1p_2}.$$

6.  $u = x_1x_2 + x_1^2$  (Final 2025 Q1). First show the preferences are strictly convex on the interior (bordered Hessian), then find the price region with an interior solution and the corner outside it. *Checkpoint:* interior iff  $p_1 > 2p_2$ , and there  $v = \frac{w^2}{4p_2(p_1 - p_2)}$ ;  $x_1 = x_2$  exactly when  $p_1 = 3p_2$ .

Then the two “do you agree” add-ons he attaches to such computations: (a) for utility 2, are  $v$  and  $e$  consistent with Props. 3.D.3 and 3.E.2 (quasiconvex, concave)? Show it directly from your formulas. (b) For utility 6, starting from  $p_1 = 3p_2$  and moving to  $(p_1 + t, p_2 - t)$ , does the consumer need more or less wealth to keep utility fixed? First by computation, then for any LNS continuous preferences (Unit 2, §2.2.2).

## 2.6 MWG exercises

Verified against your scanned copy; numbers follow the 1995 edition. ★ = closest to his style.

- **3.D.3★** HD1 utility  $\Rightarrow x$  and  $v$  are HD1 in  $w$  (straight wealth-expansion paths through the origin); converse. **Question 3.1–3.2.**
- **3.D.4★** quasilinear, strictly convex: demands for goods  $2, \dots, L$  do not depend on  $w$ ;  $v = w + \phi(p)$ ; when the nonnegativity constraint on the numeraire binds. **Questions 2 and 3.3.**
- **3.D.5★** CES with  $\alpha_1 = \alpha_2 = 1$ :  $x$  and  $v$ ; verify 3.D.2–3.D.3; limits; elasticity of substitution  $1/(1 - \rho)$ . **Midterm 2025 Q1.1** is  $\rho = \frac{1}{2}$ .
- **3.D.6** Stone–Geary (linear expenditure system): FOCs,  $x$ ,  $v$ , properties.
- **3.D.8**  $w \partial v / \partial w = -p \cdot \nabla_p v$  (Euler for a HD0 function).
- **3.E.3** existence of an EMP solution when  $p \gg 0$  and some  $x$  reaches  $u$ .
- **3.E.4** convex  $\succsim \Rightarrow h$  convex-valued; strictly convex  $\Rightarrow$  single-valued.
- **3.E.5★** HD1 utility  $\Rightarrow h(p, u) = h(p)u$  and  $e(p, u) = e(p)u$ . **Question 3.1.**
- **3.E.6★** CES Hicksian demand and expenditure function; verify 3.E.2–3.E.3.
- **3.E.7★** quasilinear  $\Rightarrow$  Hicksian demands for goods  $2, \dots, L$  do not depend on  $u$ ; form of  $e$ . **Question 3.3.**
- **3.E.8** Cobb–Douglas: verify the duality identities;  $e$  by inverting  $v$ .
- **3.E.9★** derive Prop. 3.E.2 from Prop. 3.D.3 through the duality identities, and back.

## 2.7 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|   | Result                                                                                                                    | MWG                           | Seen on         |
|---|---------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------|
| 5 | Existence (Weierstrass); convexity/uniqueness of $x(p, w)$                                                                | 3.D.1–3.D.2, p. 50–51         | —               |
| 6 | $v$ quasiconvex in $(p, w)$                                                                                               | 3.D.3, p. 56                  | Midterm 25 Q1.3 |
| 7 | UMP solution solves EMP, and back                                                                                         | 3.E.1, p. 58                  | —               |
| 8 | $e$ concave in $p$                                                                                                        | 3.E.2, p. 59                  | Final 25 Q1.5   |
| 9 | Compensated law of demand for $h$ ; HD1 $u \Rightarrow x$ , $v$ linear in $w$ ; quasilinear $\Rightarrow x_2$ free of $w$ | 3.E.4, p. 62; Ex. 3.D.3–3.D.4 | —               |

### You are here

The lecture notes end here (Lesson 7: expenditure function). Everything above is review; everything below is pre-reading. This week’s lectures finish Chapter 3: Unit 3 (3.G) first, Unit 4 (3.H) was Monday’s lab, Unit 5 (3.I) closes the chapter.

### 3 Unit 3: Shephard, Roy, Slutsky — the envelope theorem at work

**CORE. MWG 3.G. Lectures: week of October 6. Lesson 7 already stated the envelope theorem and Roy's identity; the rest is new.**

What he tests: Roy and Shephard as envelope-theorem statements, with the multiplier interpreted; the Slutsky equation and why utility maximization makes  $S$  symmetric; homothetic and Gorman special cases. Midterm 2025 Q2.1 and Final 2025 Q1.3 are this unit.

#### 3.1 Read: Chapter 3.G — relationships between demand, $v$ and $e$

##### 3.1.1 The envelope theorem

For  $V(q) = \max_x f(x; q)$  s.t.  $g(x; q) = 0$ , with Lagrangian  $\mathcal{L} = f(x; q) - \lambda g(x; q)$  and smooth interior solution  $x^*(q)$ :

$$\frac{dV(q)}{dq} = \frac{\partial \mathcal{L}(x^*(q), \lambda^*(q); q)}{\partial q}.$$

Indirect effects through  $x^*(q)$  vanish because  $x^*$  satisfies the FOCs. Interpretation of a multiplier:  $\lambda = \partial V / \partial(\text{constraint constant})$ , the shadow value of relaxing the constraint.

**Proposition 3.G.1 (Shephard's lemma)**

[MWG p. 68 · dokumen.pdf p. 88]

$u$  continuous,  $\succsim$  LNS and strictly convex:  $h(p, u) = \nabla_p e(p, u)$ .

*Two proofs (reproduce both). Envelope:*  $\mathcal{L} = p \cdot x - \mu(u(x) - u)$ , so  $\partial e / \partial p_\ell = \partial \mathcal{L} / \partial p_\ell = x_\ell$  at the optimum  $= h_\ell(p, u)$ .

*No calculus on the problem (supporting hyperplane):* fix  $\bar{p}$ ,  $\bar{x} = h(\bar{p}, u)$ . For all  $p$ ,  $\bar{x}$  is feasible, so  $\phi(p) = p \cdot \bar{x} - e(p, u) \geq 0$ , and  $\phi(\bar{p}) = 0$ . So  $\bar{p}$  minimizes  $\phi$ , and if  $e$  is differentiable,  $\nabla \phi(\bar{p}) = \bar{x} - \nabla_p e(\bar{p}, u) = 0$ . ■

Multiplier:  $\mu = \partial e / \partial u$ , the marginal cost of utility.

**Proposition 3.G.2**

[MWG p. 69 · dokumen.pdf p. 89]

If  $h$  is  $C^1$ : (i)  $D_p h(p, u) = D_p^2 e(p, u)$ ; (ii) NSD; (iii) symmetric; (iv)  $D_p h(p, u) p = 0$ .

*Proof.* (i) differentiate Shephard. (ii)  $e$  concave  $\Rightarrow$  Hessian NSD. (iii) Young's theorem on second derivatives. (iv)  $h$  HD0 in  $p \Rightarrow$  Euler. ■

**Proposition 3.G.3 (Slutsky equation)**

[MWG p. 71 · dokumen.pdf p. 91]

For  $u = v(p, w)$ :  $\frac{\partial h_\ell(p, u)}{\partial p_k} = \frac{\partial x_\ell(p, w)}{\partial p_k} + \frac{\partial x_\ell(p, w)}{\partial w} x_k(p, w)$ , i.e.  $D_p h(p, u) = S(p, w)$ .

*Proof (reproduce).* Differentiate the identity  $h_\ell(p, u) = x_\ell(p, e(p, u))$  with respect to  $p_k$ :  $\partial h_\ell / \partial p_k = \partial x_\ell / \partial p_k + (\partial x_\ell / \partial w)(\partial e / \partial p_k)$ , and  $\partial e / \partial p_k = h_k(p, u) = x_k(p, w)$  by Shephard and duality. ■

**Key corollary.** If demand comes from utility maximization,  $S(p, w) = D_p^2 e$  is **symmetric** and NSD. Without preferences one gets only negative semidefiniteness; symmetry is the content added by utility maximization.

**Proposition 3.G.4 (Roy's identity)**

[MWG p. 73 · dokumen.pdf p. 93]

If  $v$  is differentiable at  $(p, w) \gg 0$ :  $x_\ell(p, w) = -\frac{\partial v(p, w) / \partial p_\ell}{\partial v(p, w) / \partial w}$ .

Two proofs (reproduce the envelope one; Midterm 2025 Q2.1). Envelope:  $\mathcal{L} = u(x) + \lambda(w - p \cdot x)$ . Then  $\partial v / \partial w = \lambda$  and  $\partial v / \partial p_\ell = -\lambda x_\ell(p, w)$ . Divide.

Duality: differentiate  $v(p, e(p, u)) = u$  in  $p_\ell$ :  $\partial v / \partial p_\ell + (\partial v / \partial w) h_\ell(p, u) = 0$ . Evaluate at  $u = v(p, w)$ , where  $h = x$ .

Interpretation:  $\lambda$  is the marginal utility of wealth. A price rise of  $dp_\ell$  is equivalent to losing  $x_\ell dp_\ell$  dollars, worth  $\lambda x_\ell dp_\ell$  utils. ■

**Multipliers line up.** Differentiating  $e(p, v(p, w)) = w$  in  $w$ :  $\frac{\partial e}{\partial u} \cdot \frac{\partial v}{\partial w} = 1$ , so  $\mu = 1/\lambda$  at corresponding points.

**Summary diagram (memorize).**

$$\begin{array}{ccc} \text{UMP: } x(p, w), v(p, w) & \xleftrightarrow{v(p, e(p, u))=u, \quad e(p, v(p, w))=w} & \text{EMP: } h(p, u), e(p, u) \\ x = -\nabla_p v / \partial_w v \text{ (Roy)} & & h = \nabla_p e \text{ (Shephard)} \\ & D_p h = D_p x + D_w x x^\top \text{ (Slutsky)} & \end{array}$$

### 3.2 Do: Question 4 — Theory of $v$ and $e$ ; envelope theorem

**Covers:** MWG Chapter 3 (3.D–3.E, 3.G); Optimization lab (envelope theorem).

*Mimics:* Midterm 2025 Q1.3 and Q2.1; the instructor explicitly asks to “interpret the multiplier”.

Throughout,  $\succsim$  is rational, continuous and locally non-satiated on  $\mathbb{R}_+^L$ .

- 4.1 Prove that  $v(p, w)$  is quasiconvex in  $(p, w)$ , without assuming differentiability.
- 4.2 A consumer with fixed wealth  $w$  can live in town A (prices  $(1, 4)$ ), town B (prices  $(4, 1)$ ) or town C (prices  $(\frac{5}{2}, \frac{5}{2})$ ). Do you agree that, whatever the consumer’s (rational, continuous, LNS) preferences, she cannot strictly prefer C to *both* A and B? Prove. Then compute  $v$  for  $u = x_1 x_2$  and for  $u = x_1 + x_2$  and rank the three towns in each case.
- 4.3 Using the envelope theorem, prove Shephard’s lemma. Interpret the Lagrange multiplier of the EMP. How is it related to the multiplier of the UMP?
- 4.4 Do you agree that  $\frac{\partial e(p, u)}{\partial u} = \left[ \frac{\partial v(p, w)}{\partial w} \right]^{-1}$  when evaluated at  $w = e(p, u)$ ? Prove and explain.
- 4.5 Assume now that  $\succsim$  is homothetic. Do you agree that  $e(p, u) = u \cdot e(p, 1)$  and  $v(p, w) = w \cdot v(p, 1)$  (for a suitable normalization of  $u$ )? Prove. Conclude that  $v$  is of the Gorman form and write down  $a(p)$  and  $b(p)$ . What does this imply for the wealth expansion path?

### 3.3 MWG exercises

Verified against your scanned copy; numbers follow the 1995 edition. ★ = closest to his style.

- **3.G.1** Shephard’s lemma from Roy’s identity.
- **3.G.2** verify every proposition of Section 3.G for Cobb–Douglas.
- **3.G.3** linear expenditure system: Hicksian demand and  $e$ ; check 3.E.2–3.E.3 and 3.G.1–3.G.3.
- **3.G.4★** additively separable  $u$ : additivity is cardinal; no inferior goods when each  $u_\ell$  is strictly concave. **HW1 Q4 family.**
- **3.G.6** Fisher’s three-good demand system: is it generated by preferences?
- **3.G.10★** Gorman form  $v = a(p) + b(p)w$ : what  $a$  and  $b$  must satisfy for  $v$  to be an indirect utility function. **Question 3.4.**
- **3.G.11★** Gorman form  $\Rightarrow$  linear wealth-expansion curves. **Question 3.4.**
- **3.G.12★** which restrictions on the Gorman form give homothetic and quasilinear preferences.
- **3.G.15★**  $u = 2\sqrt{x_1} + 4\sqrt{x_2}$ :  $x, h, e, v$ ; verify Roy. **The closest published cousin of Midterm 2025 Q1.1.**

- **3.G.16\*** an expenditure function with parameters: restrictions for it to come from utility maximization;  $v$ ; Roy; Slutsky.
- **3.G.17** (Hausman) a local indirect utility function: recover demand and check it.

### 3.4 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|    | Result                                                                                  | MWG                   | Seen on         |
|----|-----------------------------------------------------------------------------------------|-----------------------|-----------------|
| 10 | Shephard's lemma (envelope and supporting-hyperplane proofs)                            | 3.G.1, p. 68          | Final 25 Q1.3   |
| 11 | Slutsky equation; $D_p h = S$ symmetric, NSD, $S p = 0$ ; Giffen $\Rightarrow$ inferior | 3.G.2–3.G.3, p. 69–71 | —               |
| 12 | Roy's identity (envelope proof); interpret $\lambda$ ; $\mu = 1/\lambda$                | 3.G.4, p. 73          | Midterm 25 Q2.1 |

## 4 Unit 4: Integrability — recovering preferences from demand

**SUPPORTING. MWG 3.H. Lab: Monday October 5 (integration, differential equations, recovering preferences). Pair this unit with your lab notes.**

What he tests: given a demand function, verify the conditions, write the PDE system from Shephard's lemma, solve it for  $e$ , invert to  $v$ , and recover  $u$ ; say what symmetry and negative semidefiniteness each guarantee; say in what sense the recovered utility is unique.

### 4.1 Read: Chapter 3.H — integrability

[MWG p. 75–79 · dokumen.pdf p. 95–99]

**Question:** given a demand function  $x(p, w)$ , is there a rational preference relation that generates it, and how do we recover it?

**Necessary conditions** (from §§2–3): WL, HD0,  $S(p, w)$  symmetric and NSD. **The integrability theorem:** if  $x(p, w)$  is  $C^1$  and satisfies WL, HD0 and has a symmetric NSD Slutsky matrix, then there is a rational preference relation generating it. So these conditions are also sufficient.

#### Step 1: recover $e(p, u)$ by solving PDEs

Fix  $p^0$  and normalize  $e(p^0, u) = u$  (this pins down a utility scale). Shephard plus duality give the system

$$\frac{\partial e(p, u)}{\partial p_\ell} = x_\ell(p, e(p, u)), \quad \ell = 1, \dots, L.$$

- **Why symmetry is needed (Frobenius):** a solution exists only if cross-partials agree. Differentiating,  $\partial^2 e / \partial p_k \partial p_\ell = \partial x_\ell / \partial p_k + (\partial x_\ell / \partial w) x_k = s_{\ell k}$ . Equal cross-partials  $\iff s_{\ell k} = s_{k\ell}$ .
- **Why NSD is needed:** it makes the recovered  $e$  concave in  $p$ , as an expenditure function must be.
- WL and HD0 make the recovered  $e$  HD1 in  $p$ .
- **When  $L = 2$ :** after normalizing  $p_2 = 1$  (by HD0) there is a single ODE  $de/dp_1 = x_1(p_1, 1, e)$ , which always has a solution, so symmetry imposes no restriction. Only NSD matters. With two goods there is only one independent cross-partial, so there is nothing for symmetry to constrain.

#### Step 2: recover preferences from $e$

**Proposition 3.H.1**

[MWG p. 77 · dokumen.pdf p. 97]

If  $e(p, u)$  is strictly increasing in  $u$ , continuous, increasing, HD1, concave and differentiable in  $p$ , then  $e$  is the expenditure function of the upper contour sets  $V_u = \{x \in \mathbb{R}_+^L : p \cdot x \geq e(p, u) \text{ for all } p \gg 0\}$ .

Practical shortcut:  $u(x) = \min_{p \gg 0} v(p, p \cdot x)$ , or invert  $e$  to get  $v$  and guess  $u$  from the familiar form.

#### Worked example (Cobb–Douglas)

$x_1 = \alpha w / p_1$ ,  $x_2 = (1 - \alpha)w / p_2$ . The PDEs are  $\partial \ln e / \partial p_1 = \alpha / p_1$ ,  $\partial \ln e / \partial p_2 = (1 - \alpha) / p_2$ , so  $\ln e = \alpha \ln p_1 + (1 - \alpha) \ln p_2 + c(u)$ . Choose  $c(u) = \ln u$ :  $e(p, u) = u p_1^\alpha p_2^{1-\alpha}$ ,  $v(p, w) = w / (p_1^\alpha p_2^{1-\alpha})$ . Then  $\min_{p_1} (p_1 x_1 + x_2) / p_1^\alpha$  (setting  $p_2 = 1$ ) is attained at  $p_1 = \alpha x_2 / ((1 - \alpha)x_1)$ , giving  $u(x) = x_1^\alpha x_2^{1-\alpha} / (\alpha^\alpha (1 - \alpha)^{1-\alpha})$ , a Cobb–Douglas utility as expected.

#### Exam angle

*Uniqueness:* the recovered preferences are unique; the utility function is unique only up to monotone transformation (the choice of  $c(u)$ ). Expect: “Given  $x(p, w) = \dots$ , verify the condi-

tions, find  $e$ ,  $v$ ,  $h$ , and a utility function.” Question 5 is exactly this.

## 4.2 Do: Question 5 — Integrability — recovering preferences from demand

**Covers:** MWG Chapter 3.H and 3.E–3.G; Integration/ODE lab (Oct 5).

*Mimics:* the Oct 5 lab is new and sits right before the midterm. The demand below is the instructor’s own Midterm 2025 utility in disguise — do not look it up, derive it.

A consumer’s Walrasian demand is observed to be

$$x(p, w) = \left( \frac{w p_2}{p_1(p_1 + p_2)}, \frac{w p_1}{p_2(p_1 + p_2)} \right), \quad p \gg 0, w > 0.$$

- 5.1** Verify Walras’ law and homogeneity of degree zero. Compute the Slutsky matrix and show that it is symmetric and negative semidefinite.
- 5.2** Explain precisely why the properties in 5.1 guarantee that there exists a rational, continuous, LNS preference relation generating  $x(p, w)$ . Which property is responsible for which step of the argument?
- 5.3** Use Shephard’s lemma,  $\partial e / \partial p_\ell = x_\ell(p, e(p, u))$ , to write down the system of partial differential equations that  $e(p, u)$  must satisfy. Solve it. [Hint: divide by  $e$  and work with  $\ln e$ ; note  $\frac{p_2}{p_1(p_1 + p_2)} = \frac{1}{p_1} - \frac{1}{p_1 + p_2}$ .]
- 5.4** Derive  $v(p, w)$  and  $h(p, u)$ . Then recover a utility function  $u(x_1, x_2)$  that generates this demand. [Hint: compute  $\sqrt{h_1} + \sqrt{h_2}$ .]
- 5.5** Do you agree that the utility function recovered in 5.4 is unique? Explain what uniqueness can and cannot mean here, and relate it to the constant of integration in 5.3.
- 5.6** Do you agree that the integrability argument in 5.3 works for *any*  $L$  as long as  $S$  is symmetric and negative semidefinite? What is the role of symmetry (Frobenius’ theorem) when  $L \geq 3$ , and why is it automatic when  $L = 2$ ?

## 4.3 MWG exercises

Verified against your scanned copy; numbers follow the 1995 edition.  $\star$  = closest to his style.

- **3.H.3** recover  $v$  from  $e$ .
- **3.H.5 $\star$**  given  $v$ , recover  $e$  and the direct utility function.
- **3.H.6 $\star$**   $x_\ell = \alpha_\ell w / p_\ell$  with  $\sum \alpha_\ell = 1$ : derive the expenditure function; which utility is it?
- **3.H.7 $\star$**  recover the preferences behind the demand  $x_k = w / \sum_\ell p_\ell$  (from Ex. 2.F.17), normalising  $u(1, \dots, 1) = 1$ .

## 4.4 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|    | Result                                              | MWG           | Seen on    |
|----|-----------------------------------------------------|---------------|------------|
| 13 | Integrability: PDE system; role of symmetry and NSD | 3.H, p. 75–79 | Oct. 5 lab |

## 5 Unit 5: Welfare — EV, CV, deadweight loss

**SUPPORTING. MWG 3.I, then the partial-equilibrium welfare lectures of the week of October 19 (Ch. 4, optional Ch. 10). Do parts 6.1–6.3 now, 6.4–6.5 after those lectures.**

What he tests:  $EV$  and  $CV$  as areas under Hicksian demand curves; why they coincide under quasilinearity; why a tax's deadweight loss is second order.

### 5.1 Read: Chapter 3.I — welfare evaluation

[MWG p. 80–91 · dokumen.pdf p. 100–111]

Price change  $p^0 \rightarrow p^1$ , wealth fixed at  $w$ ;  $u^0 = v(p^0, w)$ ,  $u^1 = v(p^1, w)$ .

$$EV = e(p^0, u^1) - w, \quad CV = w - e(p^1, u^0).$$

EV: the wealth change at *old* prices that is equivalent to the price change. CV: the wealth change at *new* prices that compensates for it. Both have the sign of  $u^1 - u^0$ , because  $e(\bar{p}, \cdot)$  is a **money-metric** utility for any fixed  $\bar{p}$ .

If only  $p_1$  changes (Shephard plus the fundamental theorem of calculus):

$$EV = \int_{p_1^1}^{p_1^0} h_1(s, p_{-1}, u^1) ds, \quad CV = \int_{p_1^1}^{p_1^0} h_1(s, p_{-1}, u^0) ds.$$

- Normal good:  $h_1$  increases with  $u$ . For a price *increase* ( $u^1 < u^0$ ),  $|CV| \geq |EV|$ , and the Marshallian area lies between them.
- **Quasilinear** in the numeraire: no wealth effect on good 1, so  $h_1$  does not depend on  $u$  and  $EV = CV = \Delta(\text{Marshallian consumer surplus})$ .
- Deadweight loss of a tax  $t$ :  $DWL = -EV - tx_1(p^1, w)$ , of order  $t^2$  (Harberger triangle).

### 5.2 Do: Question 6 — Welfare measurement with quasilinear utility; partial equilibrium

**Covers:** MWG Chapter 3.I and Chapter 10 (10.C–10.E); Chapter 4 (welfare).

*Mimics:* the “welfare in partial equilibrium with quasilinear utility” lectures of weeks 7–8.

Use the consumer of Question 2,  $u = x_1 + 2\sqrt{x_2}$ , with  $p_1 = 1$  normalized and the solution interior.

- 6.1** The price of good 2 rises from  $p_2^0$  to  $p_2^1 > p_2^0$ . Compute the equivalent variation and the compensating variation. Do you agree that  $EV = CV$ ? Is this specific to quasilinear preferences? Prove and explain.
- 6.2** Show that  $EV$  equals the change in the area to the left of the Hicksian demand curve for good 2, and that here it also equals the change in Marshallian consumer surplus. Why?
- 6.3** Do you agree that, for a price increase of a *normal* good,  $|CV| > |EV|$ ? Prove using the relationship between Hicksian demand curves at different utility levels, and reconcile with 6.1.
- 6.4** Instead of a price change, the government levies a per-unit tax  $t > 0$  on good 2 and returns the revenue to the consumer as a lump sum. Compute the tax revenue, the deadweight loss, and show that the deadweight loss is of second order in  $t$  (i.e.  $DWL/t \rightarrow 0$  as  $t \rightarrow 0$ ). Interpret.
- 6.5** There are  $I$  identical consumers as above and  $J$  identical price-taking firms producing good 2 with cost  $c(q) = q^2/2$  (in units of good 1). Derive the competitive equilibrium price of good 2 and quantities. Show that the equilibrium allocation maximizes Marshallian aggregate surplus, and explain why, with quasilinear utility, this is equivalent to Pareto optimality.

### 5.3 MWG exercises

Verified against your scanned copy; numbers follow the 1995 edition.  $\star$  = closest to his style.

- **3.I.3 $\star$**  a price change in one good only: compare  $CV$  and  $EV$  according to whether the good is normal or inferior.
- **3.I.5 $\star$**  quasilinear in good 1 (with  $p_1 = 1$ )  $\Rightarrow CV = EV$  for any price change. **Question 6.1.**
- **3.I.6** if aggregate  $CV > 0$ , a compensation scheme makes every consumer better off.
- **3.I.7**  $L = 3$  with a numeraire and a linear demand system: welfare measures and the integrability restrictions on the coefficients.

### 5.4 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|    | Result                                                                                             | MWG           | Seen on |
|----|----------------------------------------------------------------------------------------------------|---------------|---------|
| 14 | $EV$ , $CV$ as Hicksian areas; quasilinear $\Rightarrow EV = CV = \Delta CS$ ; DWL is second order | 3.I, p. 80–91 | —       |

## 6 Unit 6: Aggregate demand, the Gorman form, representative consumers

**CORE. MWG 4.B and 4.D. Lectures: Thursday October 15 and the week of October 19. Read this unit before October 15.**

What he tests: when aggregate demand depends only on total wealth; the planner's wealth-distribution problem, its FOC and its envelope properties; the proof that the planner's value function generates aggregate demand; what the Gorman form adds. Three of the four parts of Midterm 2025 Q2 are here.

### 6.1 Read: Chapter 4.B — aggregate demand and aggregate wealth

[MWG p. 105–109 · dokumen.pdf p. 125–129]

$I$  consumers,  $x(p, w_1, \dots, w_I) = \sum_i x_i(p, w_i)$ . When does aggregate demand depend only on  $w = \sum_i w_i$ ? For every redistribution  $dw$  with  $\sum_i dw_i = 0$  we need  $\sum_i (\partial x_{\ell i} / \partial w_i) dw_i = 0$ , which holds for all such  $dw$  iff

$$\frac{\partial x_{\ell i}(p, w_i)}{\partial w_i} = \frac{\partial x_{\ell j}(p, w_j)}{\partial w_j} \quad \text{for all } i, j, \ell, \text{ and all wealth levels.}$$

So wealth expansion paths are **straight lines, parallel across consumers**.

**Proposition 4.B.1 (Gorman)**

[MWG p. 107 · dokumen.pdf p. 127]

Consumers have parallel, straight wealth expansion paths at every  $p$  iff their indirect utilities have the **Gorman form** with a common coefficient on wealth:  $v_i(p, w_i) = a_i(p) + b(p) w_i$ .

*Proof of sufficiency (reproduce; MWG Exercise 4.B.1).* By Roy, and then summing over  $i$ ,

$$x_{\ell i}(p, w_i) = -\frac{\partial a_i / \partial p_{\ell} + (\partial b / \partial p_{\ell}) w_i}{b(p)}, \quad x_{\ell}(p, w) = -\frac{\sum_i \partial a_i / \partial p_{\ell}}{b(p)} - \frac{\partial b / \partial p_{\ell}}{b(p)} w,$$

which depends on wealth only through  $w$ . Also  $\partial x_{\ell i} / \partial w_i = -(\partial b / \partial p_{\ell}) / b$  is the same for everyone. ■

Examples: identical homothetic preferences ( $a_i = 0$ , common  $b$ ); quasilinear preferences with the same numeraire ( $b(p) = 1/p_1$ ). Remember MWG neglects boundary (nonnegativity) issues here, so the result is local.

### 6.2 Read: Chapter 4.D — representative consumers and the planner

[MWG p. 116–119 · dokumen.pdf p. 136–139]

**Definitions 4.D.1–4.D.3**

**4.D.1 Positive representative consumer:** a rational  $\succsim$  on  $\mathbb{R}_+^L$  whose Walrasian demand at  $(p, w)$  equals aggregate demand  $x(p, w)$ .

**4.D.2 Bergson–Samuelson SWF:**  $W : \mathbb{R}^I \rightarrow \mathbb{R}$ , assumed increasing, concave, differentiable when convenient.

**4.D.3 Normative representative consumer (relative to  $W$ ):** a positive representative consumer such that, for every  $(p, w)$ , wealth is distributed to solve the planner's problem (4.D.1) below, and the value function of that problem is an indirect utility function for  $\succsim$ .

**Planner's problem (4.D.1).** For given  $(p, w)$ :

$$v(p, w) = \max_{w_1, \dots, w_I} W(v_1(p, w_1), \dots, v_I(p, w_I)) \quad \text{s.t.} \quad \sum_i w_i \leq w.$$

Interior FOC with multiplier  $\lambda$ :

$$\frac{\partial W}{\partial v_i} \frac{\partial v_i}{\partial w_i} = \lambda \quad \text{for all } i. \quad (4.D.2)$$

**Interpretation:** at the optimum, the social value of an extra dollar is the same whoever receives it. MWG assumes the  $u_i$  are concave, so each  $v_i$  is concave in  $w_i$  and the FOCs are sufficient.

**Proposition 4.D.1**

[MWG p. 117 · dokumen.pdf p. 137]

If  $(w_1(p, w), \dots, w_I(p, w))$  solves (4.D.1) for every  $(p, w)$ , then the value function  $v(p, w)$  is an indirect utility function of a positive representative consumer for  $x(p, w) = \sum_i x_i(p, w_i(p, w))$ .

*Proof (reproduce; Midterm 2025 Q2.2–2.3).* Let  $x^R(p, w)$  be the demand generated by  $v(p, w)$  through Roy's identity.

*Wealth derivative.* By the envelope theorem,  $\partial v / \partial w = \lambda$ .

*Price derivative.* By the chain rule,

$$\begin{aligned} \frac{\partial v}{\partial p_\ell} &= \sum_i \frac{\partial W}{\partial v_i} \left( \frac{\partial v_i}{\partial p_\ell} + \frac{\partial v_i}{\partial w_i} \frac{\partial w_i}{\partial p_\ell} \right) \\ &= \sum_i \frac{\partial W}{\partial v_i} \frac{\partial v_i}{\partial p_\ell} + \lambda \sum_i \frac{\partial w_i}{\partial p_\ell} = \sum_i \frac{\partial W}{\partial v_i} \frac{\partial v_i}{\partial p_\ell}, \end{aligned}$$

using (4.D.2) and  $\sum_i w_i(p, w) = w$  for all  $p$  (so  $\sum_i \partial w_i / \partial p_\ell = 0$ ). This is the envelope theorem: only the direct effect survives.

*Roy.*

$$x_\ell^R = -\frac{\partial v / \partial p_\ell}{\lambda} = -\sum_i \frac{1}{\lambda} \frac{\partial W}{\partial v_i} \frac{\partial v_i}{\partial p_\ell} = -\sum_i \frac{\partial v_i / \partial p_\ell}{\partial v_i / \partial w_i} = \sum_i x_{\ell i}(p, w_i(p, w)),$$

where the middle step uses (4.D.2),  $\frac{1}{\lambda} \frac{\partial W}{\partial v_i} = \frac{1}{\partial v_i / \partial w_i}$ , and the last step is Roy for each consumer. (MWG leaves to Exercise 4.D.2 the check that  $v$  has the properties of an indirect utility function.) ■

**Interpretation of the envelope results:**  $\partial v / \partial w = \lambda$  is the social marginal value of aggregate wealth.  $\partial v / \partial p_\ell = -\lambda x_\ell(p, w)$ : a price increase costs society exactly aggregate consumption of that good, valued at the social marginal utility of wealth. This is Roy's identity at the social level.

### 6.2.1 Gorman form and the planner

**Example 4.D.2 and the result after it**

[MWG p. 119 · dokumen.pdf p. 139]

Let  $v_i(p, w_i) = a_i(p) + b(p)w_i$  (common  $b$ ). (a) With utilitarian  $W = \sum_i u_i$ , **every** wealth distribution solves (4.D.1), and  $v(p, w) = \sum_i a_i(p) + b(p)w$  gives a normative representative consumer.

(b) Stronger:  $v(p, w) = \sum_i a_i(p) + b(p)w$  is an indirect utility function of the normative representative consumer relative to **any** increasing concave  $W$ . Only the optimal distribution rule depends on  $W$ ; the representative preferences do not.

*Proof of (a); idea of (b).* (a)  $\sum_i v_i = \sum_i a_i(p) + b(p) \sum_i w_i$  depends only on  $\sum_i w_i = w$ .

(b) In utility space the feasible set is  $\{u : \sum_i u_i \leq \sum_i a_i(p) + b(p)w\}$ . At an optimum, (4.D.2) with  $\partial v_i / \partial w_i = b(p)$  forces  $\partial W / \partial u_i$  to be equal across  $i$ . If  $v(p, w) \leq v(p', w')$ , the optimal  $u$  at  $(p, w)$  has  $\sum_i u_i \leq \sum_i u'_i$ . With equal gradient components at  $u'$ ,  $\nabla W(u') \cdot (u - u') \leq 0$ , and concavity gives  $W(u) \leq W(u') + \nabla W(u') \cdot (u - u') \leq W(u')$ . So the planner's value ranks situations the same way  $v$  does. ■

**Exam angle: Midterm 2025 Q2.4, how to set it up**

Property (3) says  $W$  treats agents symmetrically *at* equal utilities: no one gets extra social weight there. The planner's FOC under Gorman is  $\frac{\partial W}{\partial u_i} b(p) = \lambda$ , so  $\nabla W \propto (1, \dots, 1)$  at the optimum.

1. The equal-utility point satisfies this FOC by property (3).
2.  $W$  is strictly concave and the feasible utility set is a half-space (convex), so the FOC is sufficient and the optimum is unique. Hence  $u_1 = \dots = u_I$ .
3. Gorman-specific? Yes: what is used is that  $\partial v_i / \partial w_i = b(p)$  is **the same for everyone**, so the utility-possibility frontier is the hyperplane  $\sum_i u_i = \text{const.}$  With different marginal utilities of wealth, (4.D.2) would require  $\partial W / \partial u_i$  to differ across  $i$ , so equal utilities generally fail.

Solving  $a_i(p) + b(p)w_i = \bar{u}$  for all  $i$  with  $\sum_i w_i = w$  gives the rule; derive it yourself and check how  $w_i$  responds to  $w$  (the answer is clean).

### 6.3 Do: Question 7 — Aggregate demand, the Gorman form and representative consumers

**Covers:** MWG Chapter 4 (4.B–4.D).

*Mimics:* Midterm 2025 Q2.2–2.4, almost part for part. Expect something of this shape to come back.

There are  $I$  consumers. Consumer  $i$  has wealth  $w_i > 0$  and preferences on  $\mathbb{R}_+^2$  represented by

$$u_i(x_{1i}, x_{2i}) = x_{1i} + \alpha_i \ln x_{2i}, \quad \alpha_i > 0.$$

Prices are  $p \gg 0$  and wealth is large enough that every solution is interior.

- 7.1 Derive  $x_i(p, w_i)$  and  $v_i(p, w_i)$ . Show that  $v_i$  is of the Gorman form  $v_i(p, w_i) = a_i(p) + b(p)w_i$  and identify  $a_i(p)$  and  $b(p)$ .
- 7.2 Do you agree that aggregate demand  $\sum_i x_i(p, w_i)$  depends on  $(w_1, \dots, w_I)$  only through total wealth  $w = \sum_i w_i$ ? Prove it for this economy. Then state and prove the general result for Gorman-form preferences, and state (without proof) whether the converse holds.
- 7.3 Find a utility function  $U(X_1, X_2)$  for a *positive* representative consumer, i.e. one whose Walrasian demand at  $(p, w)$  equals aggregate demand. Explain why its existence here does not depend on how wealth is distributed.
- 7.4 A planner chooses  $(w_1, \dots, w_I)$  with  $\sum_i w_i = w$  to maximize  $W(u_1, \dots, u_I) = \sum_i u_i$ . Do you agree that *every* wealth distribution is optimal? Prove and explain. Is the positive representative consumer from 7.3 also a normative representative consumer in this case?
- 7.5 Now let  $W$  be strictly increasing, strictly concave, and symmetric. Derive the necessary conditions for the planner's problem. Do you agree that the solution satisfies  $u_1 = u_2 = \dots = u_I$ ? Derive the optimal wealth distribution rule  $w_i(p, w)$  explicitly. How does  $w_i$  respond to an increase in total wealth  $w$ ? Is this response specific to the common  $b(p)$ ? Prove and explain.
- 7.6 Let  $V(p, w)$  be the planner's maximum value function from 7.5. Use the envelope theorem to characterize  $\partial V / \partial w$  and  $\partial V / \partial p_2$ , and interpret both.

### 6.4 Drill B: Chapter 4, the Q2 shape

**Three economies, one routine: individual  $v_i$ , aggregate demand, planner's problem, representative consumer.**

1. **Two Cobb–Douglas consumers**,  $u_i = x_{1i}^{\alpha_i} x_{2i}^{1-\alpha_i}$  with  $\alpha_1 \neq \alpha_2$ . Write  $v_i(p, w_i)$  in the form  $b_i(p)w_i$ . Show aggregate demand for good 1 depends on  $(w_1, w_2)$  and not only on  $w_1 + w_2$ . Which condition of Prop. 4.B.1 fails? What changes if  $\alpha_1 = \alpha_2$ ?
2. **Identical homothetic consumers**,  $u_i$  homogeneous of degree one, planner maximizes  $W = \sum_i \alpha_i \ln u_i$  with  $\sum_i \alpha_i = 1$ . Derive the optimal wealth-distribution rule and the planner's value function. *Checkpoint:*  $w_i(p, w) = \alpha_i w$ , independent of  $p$  (MWG Example

4.D.1, Ex. 4.D.6). Then: is the wealth rule price-independent for *any* increasing concave  $W$  here? Prove or give a counterexample.

3. **A non-Gorman pair.** Consumer 1 has  $u_1 = \sqrt{x_{11}x_{21}}$ , consumer 2 has  $u_2 = x_{12} + 2\sqrt{x_{22}}$  (the consumer of Question 2), interior solutions, planner maximizes  $W = \ln u_1 + \ln u_2$  with  $w_1 + w_2 = w$ . Write the FOC (4.D.2), solve for the rule, and say how each  $w_i$  responds to an increase in  $w$ . *Checkpoint:*  $v_1 = \frac{w_1}{2\sqrt{p_1p_2}}$ ,  $v_2 = \frac{w_2}{p_1} + \frac{p_1}{p_2}$ , and the rule is  $w_1 = \frac{1}{2}(w + p_1^2/p_2)$  (state the range of  $w$  for which it is interior). Then: is the positive representative consumer generated by this rule also normative relative to  $W$ ? Relative to the utilitarian  $W = u_1 + u_2$ ? Explain using Definition 4.D.3 and the role of a common  $b(p)$ .

## 6.5 MWG exercises

Verified against your scanned copy; numbers follow the 1995 edition. ★ = closest to his style.

- **4.B.1★** sufficiency in Prop. 4.B.1; Gorman  $v_i \Rightarrow e_i(p, u_i) = c(p)u_i + d_i(p)$ .
- **4.B.2** consumers differing by wealth and a family-size parameter: when does aggregate demand depend only on aggregates?
- **4.C.11** two quasilinear consumers,  $x_1 + 4\sqrt{x_2}$  and  $4\sqrt{x_1} + x_2$ : is aggregate demand well behaved?
- **4.D.1★** the planner distributing wealth vs. distributing bundles: same value, same allocation. **Final 2025 Q3.2–3.3 and 3.5.**
- **4.D.2★** the planner's value function has all the properties of an indirect utility function.
- **4.D.3** prove Prop. 4.D.1 allowing for corner solutions (Kuhn–Tucker).
- **4.D.6** Example 4.D.1: HD1 utilities and  $W = \sum_i \alpha_i \ln u_i \Rightarrow w_i = \alpha_i w$ .
- **4.D.7★** Example 4.D.2: utilitarian  $W$  + Gorman  $\Rightarrow$  any distribution is optimal and  $v = \sum_i a_i + bw$ . **The source of Midterm 2025 Q2.4.**
- **4.D.10** an economy with a positive but no normative representative consumer.

## 6.6 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|    | Result                                                              | MWG           | Seen on             |
|----|---------------------------------------------------------------------|---------------|---------------------|
| 15 | Gorman $\Rightarrow$ aggregate demand depends on $w$ only           | 4.B.1, p. 107 | —                   |
| 16 | Planner's value function is a positive representative consumer      | 4.D.1, p. 117 | Midterm 25 Q2.2–2.3 |
| 17 | Gorman + symmetric strictly concave $W \Rightarrow$ equal utilities | p. 119        | Midterm 25 Q2.4     |

## 7 Unit 7: The last two weeks

### 7.1 HW2 (due Sunday October 25)

#### Protocol

1. The day HW2 is posted, send me the questions. I map each to its MWG source and build a mock midterm in his format from HW2 plus the units above.
2. Do HW2 alone first, timed, before we discuss it: it is the best predictor of Q2.
3. Whatever HW2 asks about (Gorman and the planner, integrability, welfare), redo that unit's Check table from a blank page.

### 7.2 October 26–28

- **Mon Oct 26.** Midterm 2025 (ECON6000 Midterm.pdf) closed book, 80 minutes. It is entirely within scope now that Chapter 2 is gone. Grade yourself against the proofs in this pack; be harsh on every “do you agree” part that has no verdict sentence, no cited assumption, or no counterexample.
- **Tue Oct 27.** The mock built from HW2, same rules.
- **Wed Oct 28.** The full checklist below from a blank page; HW1 Q4 and the HW2 question closest to Q2 once more. Stop early. Nothing new on this day.

### 7.3 Exam-day protocol

1. Read both questions completely before writing (two minutes). Note which sub-parts are computations and which are proofs.
2. Do every computational sub-part of both questions first. They are about 40% of the paper and the proof parts often reuse their results.
3. Check each demand function you derive:  $p \cdot x(p, w) = w$  and  $x(\alpha p, \alpha w) = x(p, w)$ . If either fails, the error is upstream.
4. Then the “do you agree” parts you know best. Verdict first. If the claim is false, give an explicit counterexample with numbers. If it is true, name the proposition, state its hypotheses, and prove it.
5. Never leave a part blank: state the relevant proposition and what it would imply. He grades reasoning, and a correct partial argument earns marks.
6. Ten minutes per sub-part. If a computation is not closing by hand, it is wrong; move on and come back.

### 7.4 Master proof checklist

Tick each when you can write it on a blank page in under five minutes.

|    | Result                                                                                                                    | MWG                           | Seen on                 |
|----|---------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------|
| 1  | LNS $\Rightarrow$ Walras' law at the optimum                                                                              | 3.D.2, p. 51                  | HW1 Q3                  |
| 2  | Lexicographic preferences: no utility representation                                                                      | p. 46                         | —                       |
| 3  | Continuous + monotone $\Rightarrow$ utility (diagonal construction)                                                       | 3.C.1, p. 47                  | —                       |
| 4  | Strictly convex $\iff$ strictly quasiconcave; sums of concave functions                                                   | p. 49                         | Midterm 25 Q1.4; HW1 Q4 |
| 5  | Existence (Weierstrass); convexity/uniqueness of $x(p, w)$                                                                | 3.D.1–3.D.2, p. 50–51         | —                       |
| 6  | $v$ quasiconvex in $(p, w)$                                                                                               | 3.D.3, p. 56                  | Midterm 25 Q1.3         |
| 7  | UMP solution solves EMP, and back                                                                                         | 3.E.1, p. 58                  | —                       |
| 8  | $e$ concave in $p$                                                                                                        | 3.E.2, p. 59                  | Final 25 Q1.5           |
| 9  | Compensated law of demand for $h$ ; HD1 $u \Rightarrow x$ , $v$ linear in $w$ ; quasilinear $\Rightarrow x_2$ free of $w$ | 3.E.4, p. 62; Ex. 3.D.3–3.D.4 | —                       |
| 10 | Shephard's lemma (envelope and supporting-hyperplane proofs)                                                              | 3.G.1, p. 68                  | Final 25 Q1.3           |
| 11 | Slutsky equation; $D_p h = S$ symmetric, NSD, $S p = 0$ ; Giffen $\Rightarrow$ inferior                                   | 3.G.2–3.G.3, p. 69–71         | —                       |
| 12 | Roy's identity (envelope proof); interpret $\lambda$ ; $\mu = 1/\lambda$                                                  | 3.G.4, p. 73                  | Midterm 25 Q2.1         |
| 13 | Integrability: PDE system; role of symmetry and NSD                                                                       | 3.H, p. 75–79                 | Oct. 5 lab              |
| 14 | $EV$ , $CV$ as Hicksian areas; quasilinear $\Rightarrow EV = CV = \Delta CS$ ; DWL is second order                        | 3.I, p. 80–91                 | —                       |
| 15 | Gorman $\Rightarrow$ aggregate demand depends on $w$ only                                                                 | 4.B.1, p. 107                 | —                       |
| 16 | Planner's value function is a positive representative consumer                                                            | 4.D.1, p. 117                 | Midterm 25 Q2.2–2.3     |
| 17 | Gorman + symmetric strictly concave $W \Rightarrow$ equal utilities                                                       | p. 119                        | Midterm 25 Q2.4         |
| 18 | Engel aggregation $\Rightarrow$ not all goods inferior; increasing differences $\Rightarrow$ monotone argmax              | 2.E.3; lab                    | —                       |

## A Good to know: comparative statics and partial-equilibrium welfare

**GOOD TO KNOW.** Comparative statics lab (Monday October 19) and the welfare lectures of that week (Ch. 4, optional Ch. 10). Syllabus topics: monotone comparative statics and the envelope theorem without differentiability.

What he tests: signing derivatives of demand through the FOCs and the Slutsky equation; proofs that need no differentiability; the surplus logic of quasilinear markets.

### A.1 Read: comparative statics of the consumer problem

#### A.1.1 Differentiating the first-order conditions

For  $L = 2$ , an interior solution and  $u$  twice continuously differentiable, the FOCs are  $u_1(x) - \lambda p_1 = 0$ ,  $u_2(x) - \lambda p_2 = 0$ ,  $w - p \cdot x = 0$ . Differentiating with respect to  $w$  gives the bordered-Hessian system

$$\begin{pmatrix} u_{11} & u_{12} & -p_1 \\ u_{21} & u_{22} & -p_2 \\ -p_1 & -p_2 & 0 \end{pmatrix} \begin{pmatrix} \partial x_1 / \partial w \\ \partial x_2 / \partial w \\ \partial \lambda / \partial w \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix},$$

which the implicit function theorem solves whenever the bordered Hessian is nonsingular (the same condition under which  $x(p, w)$  is differentiable, MWG p. 94–95). The last row is Engel aggregation,  $p \cdot \partial x / \partial w = 1$ : **not all goods can be inferior**. Differentiating with respect to  $p_k$  instead and using the Slutsky equation  $\partial x_\ell / \partial p_k = s_{\ell k} - x_k \partial x_\ell / \partial w$  gives the sign results:  $s_{\ell \ell} \leq 0$  always, so a Giffen good must be inferior; an inferior good need not be Giffen.

#### A.1.2 Monotone comparative statics (no derivatives)

**Increasing differences  $\Rightarrow$  monotone maximizers**

Let  $\phi : X \times T \rightarrow \mathbb{R}$  with  $X, T \subset \mathbb{R}$  have *increasing differences*:  $\phi(x', t') - \phi(x, t') \geq \phi(x', t) - \phi(x, t)$  whenever  $x' > x$  and  $t' > t$ . If  $x^*(t)$  is the unique maximizer of  $\phi(\cdot, t)$  on  $X$ , then  $x^*$  is nondecreasing in  $t$ .

*Proof (reproduce).* Let  $t' > t$  and suppose, for a contradiction,  $a = x^*(t) > b = x^*(t')$ . Uniqueness at  $t$  gives  $\phi(a, t) > \phi(b, t)$ . Increasing differences with  $x' = a > x = b$  gives  $\phi(a, t') - \phi(b, t') \geq \phi(a, t) - \phi(b, t) > 0$ , so  $a$  beats  $b$  at  $t'$ , contradicting the optimality of  $b$  at  $t'$ . Without uniqueness the same argument shows that the largest and the smallest maximizers are nondecreasing. ■

Application: for  $u = x_1 + f(x_2)$  with  $f$  strictly increasing and strictly concave (not necessarily differentiable),  $x_2$  solves  $\max_{x_2} f(x_2) - (p_2/p_1)x_2$ . With  $t = -p_2/p_1$  and  $\phi(x, t) = f(x) + tx$ ,  $\phi(x', t') - \phi(x, t') - [\phi(x', t) - \phi(x, t)] = (t' - t)(x' - x) \geq 0$ . Hence  $x_2$  is nonincreasing in  $p_2/p_1$ , with no concavity, differentiability or interiority used. This is the “prove it without differentiability” style of Midterm 2025 Q1.4.

#### A.1.3 Partial equilibrium with quasilinear utility (Chapter 10, essentials)

$I$  consumers with  $u_i = m_i + \phi_i(x_i)$  ( $m_i$  the numeraire,  $\phi_i$  concave),  $J$  firms producing the good from the numeraire at cost  $c_j(q_j)$  (convex). A competitive equilibrium price  $p^*$  satisfies  $\phi'_i(x_i) = p^* = c'_j(q_j)$  for all  $i, j$  and  $\sum_i x_i = \sum_j q_j$ .

- **Marshallian aggregate surplus**  $S = \sum_i \phi_i(x_i) - \sum_j c_j(q_j)$ . The equilibrium allocation maximizes  $S$  subject to feasibility: the FOCs of the two problems coincide. With quasilinear utility, utility is transferable one-for-one through the numeraire, so the Pareto optimal allocations are exactly the surplus maximizers (any distribution of the numeraire). This is the partial-equilibrium version of the welfare theorems (MWG 10.D).

- **A per-unit tax**  $t$  drives a wedge  $p_{\text{consumer}} = p_{\text{firm}} + t$ . The lost surplus (deadweight loss) is the area between the demand and supply curves over the quantity reduction, approximately  $\frac{1}{2}t^2 |dx/dt|$ : second order in  $t$ , so a small tax costs little and the loss grows with the square of the rate. Question 6.4 computes this exactly for one consumer.

## A.2 Do: Question 8 — Comparative statics

**Covers:** MWG 3.D–3.G (Slutsky equation, implicit function theorem); Comparative statics lab (Oct 19); monotone comparative statics (syllabus topic).

*Mimics:* the lab material; the “prove without differentiability” style of Midterm 2025 Q1.4.

- 8.1 For  $L = 2$  and an interior solution of the UMP with  $u$  twice continuously differentiable, write the first-order conditions and apply the implicit function theorem to obtain the linear system that determines  $(\partial x_1/\partial w, \partial x_2/\partial w, \partial \lambda/\partial w)$ . Do you agree that both goods cannot be inferior at the same time? Prove (there is a one-line proof that needs no matrix algebra).
- 8.2 Do you agree that a Giffen good must be inferior? Do you agree that an inferior good must be Giffen? Prove and explain using the Slutsky equation.
- 8.3 Let  $u(x_1, x_2) = x_1 + f(x_2)$  with  $f$  strictly increasing and strictly concave, *not necessarily differentiable*, and let the solution be interior with  $x_2^*(p)$  unique. Prove that  $x_2^*$  is non-increasing in  $p_2/p_1$  *without using derivatives*. [Hint: compare the objective at  $x_2^*(p)$  and  $x_2^*(p')$  under both price ratios and add the two inequalities. This is a monotone comparative statics argument.]
- 8.4 Generalize 8.3: let  $\phi(x, t)$  be a function on  $\mathbb{R} \times \mathbb{R}$  with *increasing differences*, i.e.  $\phi(x', t') - \phi(x, t') \geq \phi(x', t) - \phi(x, t)$  whenever  $x' > x$  and  $t' > t$ . Let  $x^*(t)$  be the (assumed unique) maximizer of  $\phi(\cdot, t)$  on a fixed set  $X \subset \mathbb{R}$ . Prove that  $x^*(t)$  is non-decreasing in  $t$ . Which assumptions that you usually make (concavity, differentiability, interiority) were *not* needed?

## A.3 MWG exercises

Chapter 10 is optional in the syllabus; do these only if everything in Units 1–6 is solid.

Verified against your scanned copy; numbers follow the 1995 edition. ★ = closest to his style.

- **10.C.4** a central authority allocating a fixed quantity among quasilinear consumers: characterize the optimum.
- **10.C.5★** comparative statics of a tax by applying the implicit function theorem to the equilibrium system.
- **10.C.6** specific vs ad valorem taxes: equivalence and incidence.

## A.4 Check

Before leaving this unit, write each of these on a blank page in under five minutes.

|    | Result                                                                                                       | MWG        | Seen on |
|----|--------------------------------------------------------------------------------------------------------------|------------|---------|
| 18 | Engel aggregation $\Rightarrow$ not all goods inferior; increasing differences $\Rightarrow$ monotone argmax | 2.E.3; lab | —       |